BUrl: The Bucknell URL Shortener

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Abstract

BUrl is a custom URL shortening platform made exclusively for Bucknellians to address the limitations of third-party services. BUrl delivers an intuitive experience and functional tools to make link management effortless. Furthermore, the platform provides insightful data to bring purpose to the created links. BUrl was created following a agile methodology to optimize scalability and sustain maintainability of the website, keeping the users' needs in mind. Initial user feedback and automated testing indicates high satisfaction and effective performance establishing BUrl as a valuable tool for the Bucknell community.

Motivation

Current link shortening tools such as TinyURL and Bitly, are expensive solutions for users who desire to have branded links and easily accessible analytics of these links. Our client requested a professional Bucknell branded shortener and a full-service links dashboard:

*"I communicate with a number of audiences across a number of digital platforms. It **doesn't look very professional** to use a widely used and available URL shortener option. . ."*

*"I also currently have to pull analytics from a number of different platforms . . . **forcing me to combine analytics** to get a true representation of the interest in the link rather than having a 'one-stop-shop.'" [1]*

Initial Goals

1. Centralized solution for managing link distribution and engagement tracking.
2. Exclusive Bucknell University branded URL shortener.
3. Intuitive User-Interface with >80% user satisfaction.
4. Exceeds features of existing platforms.

Final Features

- Link Analytics and Data Visualizations
- Google Authentication
- AI-Generated Link Suggestions
- QR Code Generator
- Free Chrome Extension

System Design

Technical Framework

- Used the Phoenix Framework, built on the Elixir programming language for:
  - Scalability
  - Modern web-application development
  - Robust architecture
- Used PostgreSQL due to compatibility with Phoenix/Elixir.
- Deployed on Fly.io.

Phoenix Elixir PostgreSQL

System Design

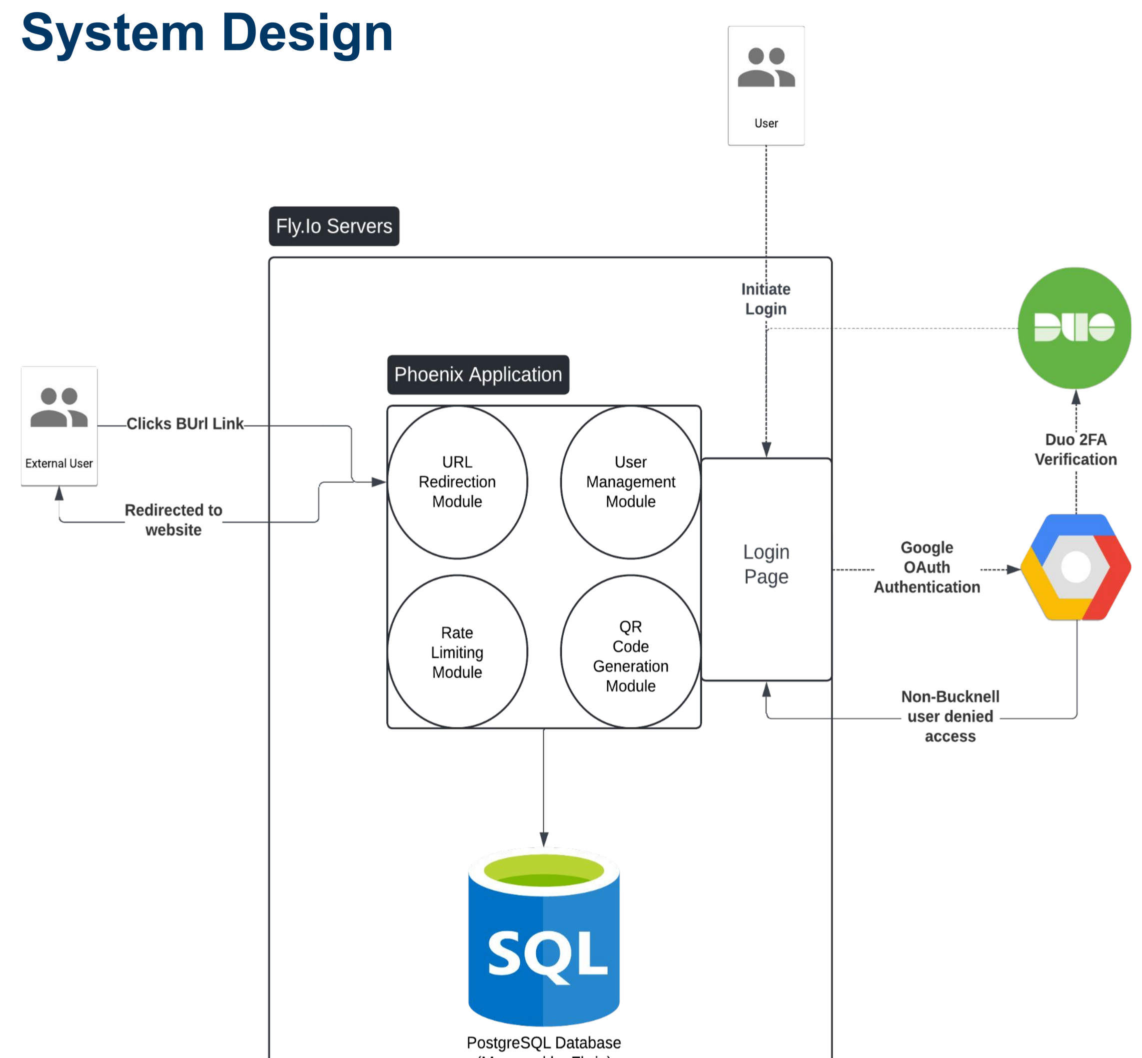
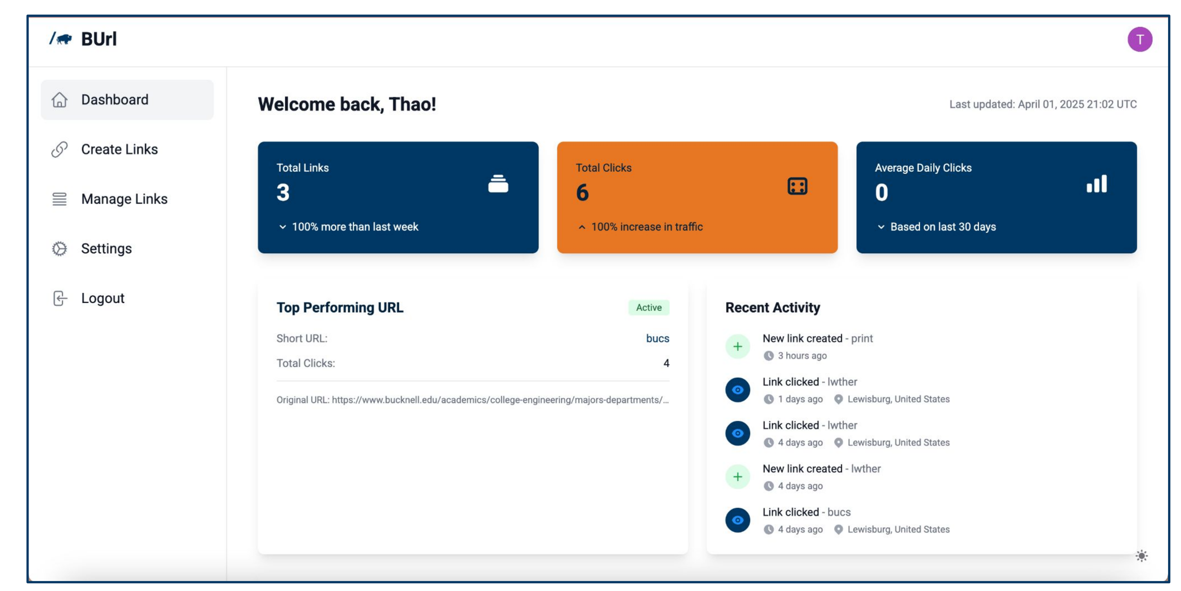
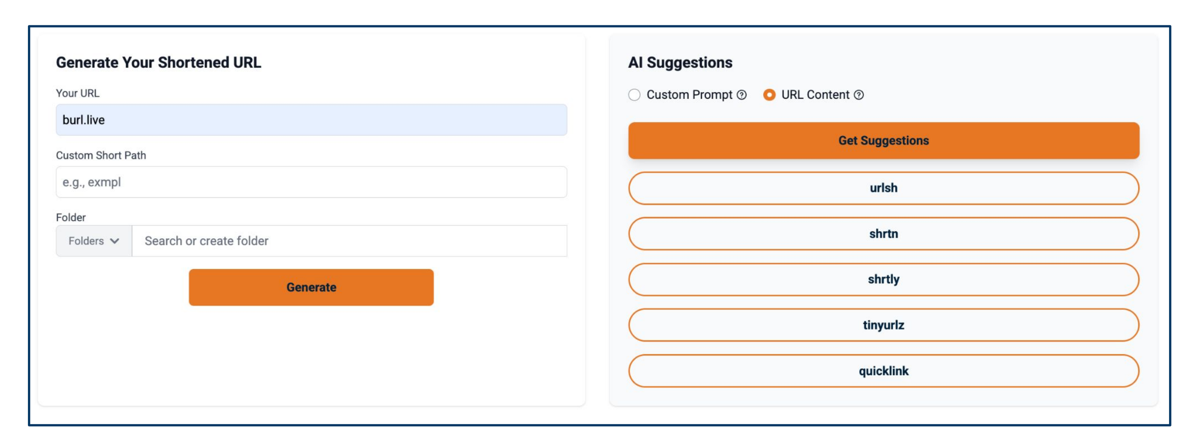


Diagram of the systems flow including the redirect, scope of server and application, and user authentication method

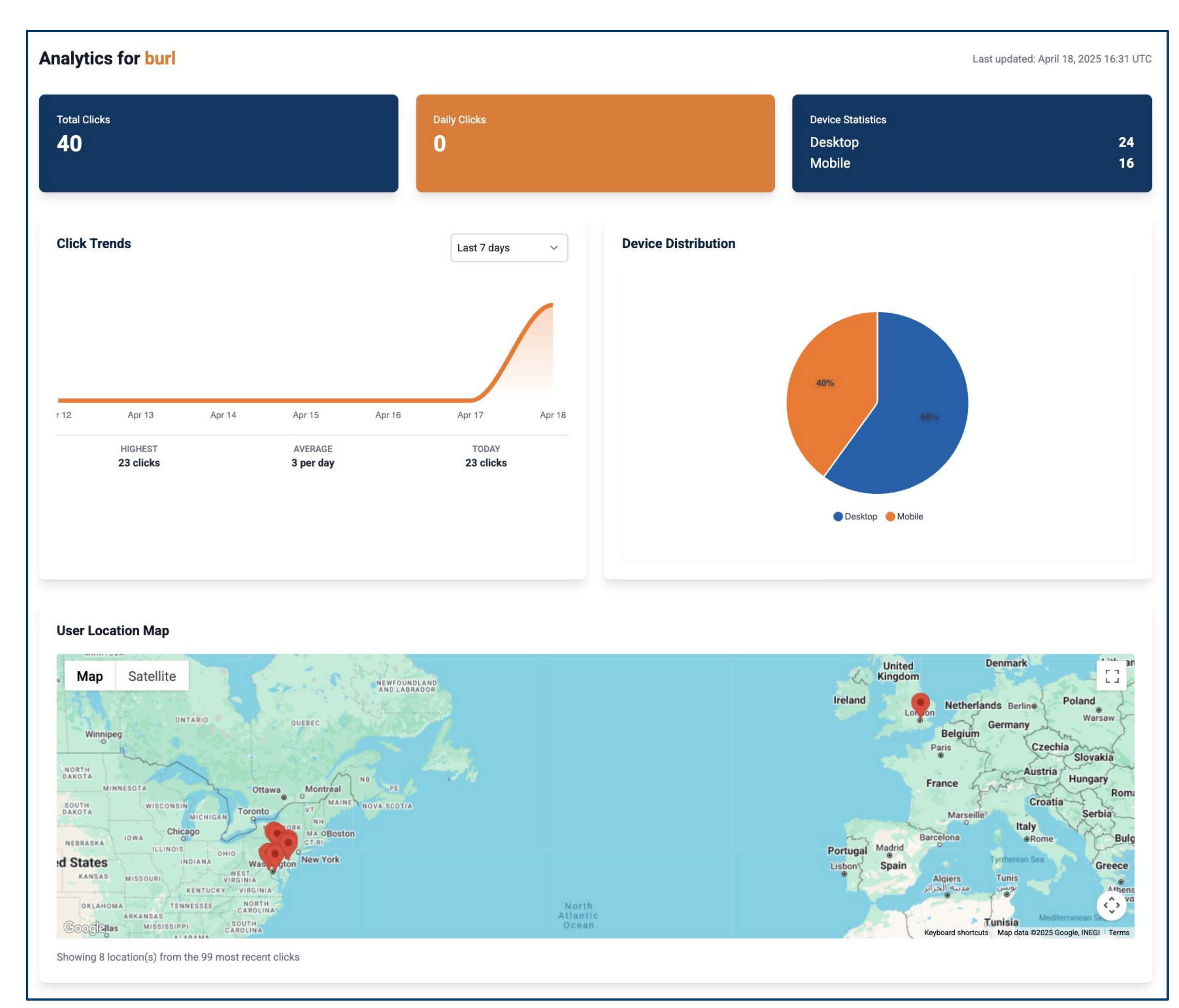
Final Product



BUrl's live "Dashboard" Page



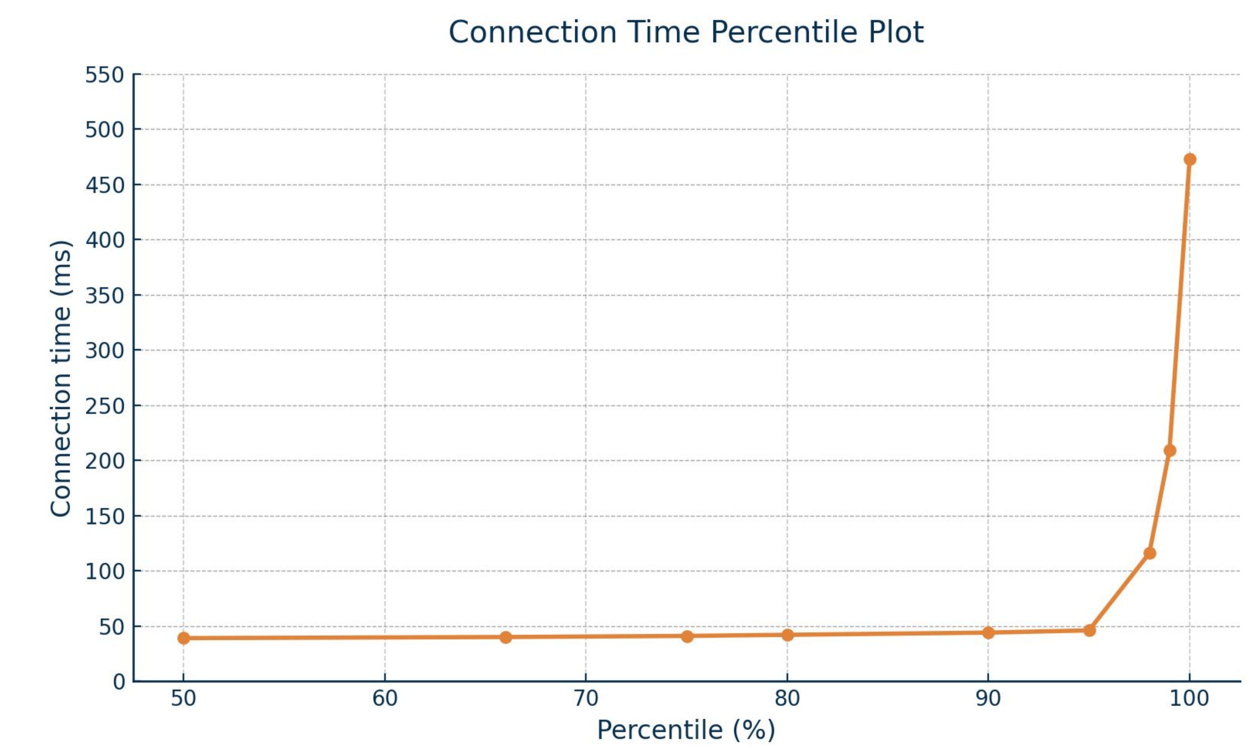
BUrl's live "Create Link" Page



Screenshots of a BUrl link's analytics

Performance

Short Link Connection Times (ms)



Connection-time percentiles from the 1,000 ApacheBench runs on Bucknell's network (302-redirect latency excluded).
Min = 33 ms; mean = 44 ms; max = 473 ms.

General Performance

Total Users:	Total Links:	Total Clicks:
30	62	1995

94% User intuitiveness Score

Glossary

1. PostgreSQL- *A powerful, free database system that helps apps store and organize information.*
2. Phoenix Framework- *A tool for building fast, interactive web apps, especially in real time.*
3. Fly.io- *A service that helps run apps closer to users, so they load faster everywhere in the world.*

References

1. Merriett, Todd. "Introduction Information/Quote." Received by Will Choi, 10/2/24.